

VeritasCarbon: Appendix Materials

Supplementary Documentation for VLDB 2027 Submission

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Dataset as a Core Contribution

In addition to the CoDE multi-agent framework, this paper makes a substantial dataset contribution by releasing **VeritasCarbon-ESG-35K**, the largest publicly available traceable ESG instruction dataset to date. The dataset contains 35,009 high-quality instruction–response pairs generated from 17,721 real-world ESG disclosure documents spanning four corpus layers (domain textbooks, CSR reports, regulatory guidelines, and industry analyses). Every pair is fully traceable to its source document chunk and includes rich generation metadata (quality score, selected experts, collaboration mode, knowledge injection count), enabling fine-grained analysis of data provenance and quality.

Unlike existing ESG corpora that are limited to classification labels or short extracts, VeritasCarbon-ESG-35K provides *task-oriented instructions* with diverse types (Q&A, summarization, extraction, classification, analysis, temporal analysis, benchmark comparison, greenwashing detection, standard alignment, knowledge graph construction, and consistency verification). This makes it directly usable for instruction tuning of large language models in the ESG and sustainable-finance domains.

The dataset is released under a tiered strategy to balance accessibility and copyright constraints:

- A representative 2,000-pair sample is included in the GitHub repository for immediate inspection and replication of the main evaluation (Table 2);
- The full 35,009-pair dataset is hosted on Hugging Face Datasets (<https://huggingface.co/datasets/Yihan-JIANG/VeritasCarbon-ESG-35K>) under CC BY-SA 4.0;
- Complete provenance metadata for all 17,721 source documents is provided in the repository, and the full dataset can be reproduced from source using the provided notebooks.

A.1 Data and Code Availability

Code. The complete VeritasCarbon framework, evaluation scripts, and reproducibility instructions are available at <https://github.com/YihanJIANG-lab/VeritasCarbon-VLDB2027> under the MIT License.

Sample Data. A representative 2,000-pair sample (`veritascarbon_sample_2000.jsonl`, 8.7 MB) is included in the GitHub repository. It is drawn with `random.seed(42)` from the full 35,009-pair pool and matches the evaluation protocol used in Table 2.

Full Dataset. The complete VeritasCarbon-ESG-35K dataset (35,009 instruction–response pairs, ~153 MB) is hosted on Hugging Face Datasets:

<https://huggingface.co/datasets/Yihan-JIANG/VeritasCarbon-ESG-35K>

It can be loaded in one line:

```
from datasets import load_dataset
ds = load_dataset("Yihan-JIANG/VeritasCarbon-ESG-35K", split="train")
```

Raw Corpus. The source ESG documents (~2.7 GB, 17,721 files) contain copyrighted material and cannot be redistributed in full. Complete provenance metadata (document IDs, titles, source URLs, and chunk-to-document mappings) is provided in `data/raw_corpus/CORPUS_MANIFEST.md`.

A.2 Reproducibility Checklist

Table 1 provides a 30-minute verification checklist for reviewers. All commands run on CPU; no GPU is required for verification.

Table 1: 30-Minute Reproducibility Checklist

Claim	Verification Command	Expected Output	Time
Table 2: CoDE ROUGE-L	<code>cat results/outputs/intrinsic_comparison.json</code>	0.8380	30 s
Table 2: CoDE BLEU-4	same file	0.1932	—
Table 2: Domain Rel.	same file	0.3637	—
Table 2: FactCheck	same file	0.9478	—
Table 3: Ablation	<code>cat results/figures_and_tables/table3_ablation.tex</code>	ablation	30 s
Fig 5: Quality dist.	<code>python scripts/generate_paper_results.py</code>	PDF/PNG files	2 min
Dataset size: 35,009	<code>cat results/figures_and_tables/table2_data_statistics.json</code>	35,009	10 s
Provenance: traceable	Inspect any record in <code>source_chunk_id</code> <code>data/sample/*.jsonl</code>	present	1 min
Evaluation pipeline	Run <code>intrinsic_evaluation_03_03.py</code> on sample	metrics JSON	10 min

Notes.

- Tracks A–B (rows 1–8) require only the GitHub repository (~10 MB download).
- Track C (full regeneration) requires the raw corpus (~2.7 GB) and an A100 GPU; see `REPRODUCIBILITY.md` for details.
- All random sampling uses `random.seed(42)`.

A.3 Prompt Templates

The CoDE framework employs 11 specialized expert agents plus a MetaExpert orchestrator. Each expert receives a system prompt that defines its persona, task, and output format. All prompts support two modes:

- **Default mode:** The expert generates a task directly from the text chunk.
- **Dynamic mode:** The expert receives an additional work instruction from the MetaExpert and must align its output to that instruction.

The dynamic mode simply prepends `Work instruction: {dynamic_instruction}` to the requirements block; the remainder of the prompt is identical. Below we present the default-mode prompts for representative experts, followed by a summary table for all 11 experts, and finally the MetaExpert orchestration prompt.

A.3.1 Q&A Expert

```
You are a senior ESG (environment, social, governance) Q&A generation expert.
Task: Generate one high-quality Q&A pair from the given text.
Requirements:
1. The question should be highly relevant to the text.
2. The answer must be findable in the text or reasonably inferred.
3. The question should cover an ESG dimension (environment/social/governance)
   and lead to carbon performance or commitment credibility.
4. The question should be specific and clear.
5. The answer should be accurate and complete.
{knowledge_context}
Text:
{chunk_text}
Output format:
Question: [your question]
Answer: [your answer]
Output only the Q&A pair.
```

A.3.2 Analysis Expert

```
You are a senior ESG analysis expert.
Task: Generate one analysis task (instruction-answer pair) from the text.
Requirements:
1. The instruction should ask to analyze ESG content (e.g. comparison,
   trend, impact) and lead to carbon/commitment assessment.
2. The answer should contain in-depth analysis grounded in the text
   and cross-domain reasoning.
3. Be critical and assess credibility of carbon commitments.
{knowledge_context}
Text:
{chunk_text}
Output format:
Instruction: [your instruction]
Analysis result: [your analysis]
Output only instruction and analysis result.
```

A.3.3 Greenwashing Detection Expert

```
You are a senior ESG greenwashing detection expert, skilled at identifying
misleading or exaggerated claims in reports.
Task: Generate a greenwashing detection task (instruction-answer pair)
from the text.
Requirements:
1. Identify greenwashing patterns (refer to common patterns list).
```

```

2. Quantify greenwashing risk (0-1, 1 = highest risk).
3. Provide evidence chain (quote source text).
4. Produce greenwashing risk report.

Common greenwashing patterns:
1. Vague commitments: "committed to", "plan to" without concrete targets
2. Selective disclosure: only favorable data, hiding unfavorable info
3. Exaggerated impact: overstating environmental project effects
4. Time mismatch: presenting future plans as achieved results
5. Scope confusion: mixing scope 1/2/3 emissions
6. Inconsistency: commitments vs actual actions
7. Lack of verification: no third-party or traceable evidence
{knowledge_context}
Text:
{chunk_text}
Output format:
Instruction: [detection instruction: identify greenwashing patterns]
Answer: [detection result: matched patterns, risk level, evidence chain]
Output only instruction and answer.

```

A.3.4 Consistency Verification Expert

```

You are a senior ESG consistency verification expert, skilled at detecting
data contradictions and gaps in reports.
Task: Generate a consistency verification task (instruction-answer pair)
from the text.
Requirements:
1. Check whether the same data is stated consistently across the text.
2. Identify contradictions (e.g. scope-1 emissions differing across sections).
3. Detect "missing data" patterns (only favorable data disclosed).
4. Produce consistency report.
{knowledge_context}
Text:
{chunk_text}
Output format:
Instruction: [verification instruction: check data consistency]
Answer: [verification result: consistency assessment, contradictions,
         missing patterns]
Output only instruction and answer.

```

A.3.5 Summary of All 11 Experts

Table 2 summarizes the role and output format of each expert agent. All experts share the same prompt structure (persona + task + requirements + knowledge context + chunk text + output format); only the specific task description and output labels differ.

A.3.6 MetaExpert Orchestrator

The MetaExpert designs work instructions for subordinate experts, evaluates their outputs, and provides iterative feedback. Its prompt dynamically incorporates the target expert type, core topics, chunk preview, and collaboration context.

```

# Role
You are a top carbon-strategy analyst. Your task is to evaluate all
information related to the company's carbon emissions and carbon-reduction
commitments. Now design one work instruction for a subordinate analyst.

# Context
- Subordinate role: {expert_type} ({expert_desc})
- Chunk is from report section [{section_name}]
- Core topics: {topics_str}

```

Table 2: CoDE Expert Agent Summary

Expert	Core Task	Output Labels
Q&A Expert	Generate a specific question-answer pair grounded in the text.	Question / Answer
Summary Expert	Generate a summarization task (100–200 words) highlighting ESG content.	Instruction / Summary
Extraction Expert	Extract structured ESG information (emissions, employee count, etc.).	Instruction / Extraction result
Classification Expert	Classify text by ESG dimension (E/S/G/Mixed) with rationale.	Instruction / Classification result
Analysis Expert	Provide in-depth cross-domain analysis and critical assessment.	Instruction / Analysis result
Temporal Analysis Expert	Analyze multi-year trends, commitment consistency, and data gaps.	Instruction / Answer
Benchmark Comparison Expert	Compare company data against industry averages and flag anomalies.	Instruction / Answer
Greenwashing Detection Expert	Identify misleading claims and quantify greenwashing risk.	Instruction / Answer
Standard Alignment Expert	Identify followed standards (GRI, TCFD, SASB, etc.) and compliance gaps.	Instruction / Answer
Knowledge Graph Expert	Extract entities and relations; detect broken commitment-action links.	Instruction / Answer
Consistency Verification Expert	Detect data contradictions and selective disclosure patterns.	Instruction / Answer

```

- Preview: {chunk_text[:500]}...
- Collaboration: This run uses multiple experts: {all_experts}
  Current expert: {expert_type}; Others: {other_experts}
  Requirement: Your instruction should match {expert_type}'s role and
  complement others (avoid duplication).

# Task
Generate one work instruction for this analyst. The instruction must:
1. Carbon-centric: lead to assessing carbon performance or commitment credibility.
2. Cross-domain: use information in this chunk as evidence for carbon strategy.
3. Depth: guide the analyst to compare, evaluate, predict, or critique
   -- not just extract.

# Output
Work instruction:

```

Feedback round. After receiving the expert’s output, the MetaExpert runs a quality-assessment prompt (not shown for brevity) that scores correctness, completeness, relevance, and structure on a 0–1 scale. If any dimension falls below threshold, a refined work instruction is generated and the expert re-executes, up to $R = 2$ rounds.

A.4 Hyperparameters

Table 3 lists all fixed hyperparameters used in the experiments. All random sampling steps use `random.seed(42)`.

Table 3: Complete Hyperparameter Configuration

Parameter	Value	Source File
Base model	Qwen2-72B-Instruct	<code>configs/config.yaml</code>
Quantization	4-bit (NF4)	Unsloth default
Temperature	0.7	<code>coe_framework_02_03.py</code>
Max new tokens (instruction)	512	—
Max new tokens (response)	1,024	—
Chunk size	2,000 chars	<code>text_chunker_01_03.py</code>
Chunk overlap	200 chars	<code>text_chunker_01_03.py</code>
Expert selection threshold	0.5	<code>expert_selector_02_01.py</code>
Collaboration mode	parallel / sequential	<code>coe_framework_02_03.py</code>
Feedback rounds (R)	2 (default)	<code>meta_expert_02_09.py</code>
Quality threshold (accept)	0.6	<code>meta_expert_02_09.py</code>
Knowledge items per chunk	up to 5	<code>domain_knowledge_02_02.py</code>
Evaluation sample size	2,000 pairs	<code>intrinsic_evaluation_03_03.py</code>
Random seed	42	All sampling steps

A.5 Expert Selection Logic

Expert selection is performed by a rule-based feature extractor (`expert_selector_02_01.py`) that scores each chunk against predefined feature patterns. No LLM is invoked during selection; the process is fully deterministic.

1. **Feature extraction.** The chunk is analyzed for:
 - Keyword presence (carbon, emission, energy, employee, training, governance, etc.)
 - Numerical data density (counts, percentages, monetary values)
 - Temporal markers (years, quarters, “since”, “by 20xx”)
 - Standard references (GRI, TCFD, SASB, ISO 14001, CDP)
 - Comparative language (“higher than”, “below average”, “ranked”)
 - Entity types (company names, plant locations, regulatory bodies)
2. **Expert scoring.** Each of the 11 expert types defines a scoring function $s_i(chunk) \in [0, 1]$ based on its target features. For example:
 - *TemporalAnalysisExpert*: high score if temporal markers ≥ 2 and numerical trends present.
 - *GreenwashingDetectionExpert*: high score if vague-commitment keywords + selective-disclosure patterns detected.
 - *KnowledgeGraphExpert*: high score if entity count ≥ 5 and relation verbs present.
3. **Thresholding.** Experts with $s_i \geq 0.5$ are selected. If no expert meets the threshold, the fallback *QAEExpert* is assigned.
4. **Collaboration mode.** If $|\{s_i \geq 0.5\}| \geq 2$, parallel collaboration is used; otherwise sequential.

MetaExpert override. After expert selection, the MetaExpert may dynamically adjust the assignment based on core topics extracted from the chunk. This override is rare ($< 5\%$ of chunks) and logged in metadata.

A.6 Error Analysis

To illustrate the quality spectrum of VeritasCarbon-ESG-35K, we present representative examples drawn from the 2,000-pair sample. The quality score is a composite metric (correctness 0.4, completeness 0.3, relevance 0.2, structure 0.1) assigned by the MetaExpert.

A.6.1 High-Quality Example (Score 0.915)

Assigned experts: Summary Expert + Consistency Verification Expert

Instruction: Generate a summary about the 2024–2025 A-share ESG fund issuance trends, listed companies’ information disclosure quality, carbon-risk considerations in investment decisions, corporate carbon-emission management progress, ESG evaluation-system effectiveness, and the potential of China’s ESG market to promote green transformation.

Answer: Since 2024, A-share ESG public fund issuance has reached approximately 216 billion yuan. Although the issuance pace has slowed, 18 new funds were still issued from Q4 to Q3 of the following year. The industry’s overall scale has rebounded against the trend, showing a structural shift from quantity expansion to quality improvement. Under the dual-wheel drive of policy and market, China’s ESG market is gradually maturing, with carbon-risk considerations becoming an important factor in investment decisions.

Why it scores high: The instruction demands multi-faceted synthesis (funds, disclosure, carbon risk, evaluation systems), and the answer integrates quantitative evidence (216B yuan, 18 funds) with qualitative assessment (structural shift, maturing market). The Consistency Verification Expert cross-checks that fund-scale figures are internally coherent.

A.6.2 Low-Quality Example (Score 0.000)

Assigned experts: Summary Expert

Instruction: Generate a summary for the following text.

Answer: This article aims to analyze how ESG information disclosure affects employees’ green innovation efficiency and its role in promoting companies’ real carbon-reduction commitments. Special attention is paid to identifying and criticizing potential "greenwashing" behavior, i.e., phenomena where companies superficially promote environmental measures without taking substantive action. In addition, the article...

Why it scores low: The output is a generic abstract of the source text rather than a task-oriented instruction–response pair. The “instruction” is merely “generate a summary,” and the “answer” restates the paper’s purpose without concrete data or analysis. This pattern occurs when the MetaExpert’s work instruction is too vague and no second expert is assigned to refine depth.

A.6.3 Key Failure Patterns

Table 4 summarizes the most common failure modes observed in the bottom-decile records.

Table 4: Common Failure Patterns in Low-Quality Records

Pattern	Description	Typical Cause
Generic summary	Answer restates source abstract without task specificity.	Single Summary Expert, no dynamic instruction.
Vague instruction	Instruction is a broad request (“analyze ESG”).	MetaExpert topics too generic.
Missing evidence	Claims lack concrete numbers or source citations.	No Extraction/Benchmark Expert assigned.
Self-reference	Answer refers to “this article” or “the text.”	Expert fails to rephrase into third-person response.

A.7 Dataset Schema

Each record in the dataset is a JSON object with the following schema. All fields are guaranteed present; optional sub-fields within `metadata` default to `null` or empty lists when unavailable.

Table 5: VeritasCarbon-ESG-35K JSON Schema

Field	Type	Description
<code>instruction</code>	string	The generated task instruction (Chinese, avg 106.5 chars).
<code>input</code>	string	Source text chunk from the ESG document (up to ~2,000 chars).
<code>output</code>	string	Generated response (Chinese/English, avg 380.4 chars).
<code>source_chunk_id</code>	string	Unique chunk identifier (e.g., <code>Layer2_CSR_2024_000920_chunk_21</code>).
<code>metadata</code>	object	Generation provenance and quality metadata (see below).

Metadata Object

Field	Type	Description
<code>quality_score</code>	float	Composite score in $[0, 1]$ (mean 0.667, std 0.103).
<code>selected_experts</code>	list[str]	IDs of expert agents invoked (e.g., <code>[summary_expert, consistency_verification_expert]</code>).
<code>expert_names</code>	string	Human-readable expert names.
<code>selection_reasons</code>	object	Feature scores that triggered each expert’s selection.
<code>use_meta_expert</code>	bool	Whether MetaExpert orchestration was used.
<code>dynamic_instructions</code>	string	The work instruction generated by MetaExpert (if any).
<code>knowledge_items_count</code>	int	Number of injected domain-knowledge items (0–5).
<code>collaboration_mode</code>	string	<code>parallel</code> or <code>sequential</code> .
<code>max_iterations</code>	int	Feedback rounds executed (0–2).
<code>model</code>	string	Generator model ID (<code>Qwen2-72B-Instruct</code>).
<code>framework</code>	string	Always <code>CoDE</code> .

Quality score composition. The MetaExpert evaluates four dimensions, each scored 0–1:

- **Correctness** (weight 0.4): Factual accuracy against the source chunk.
- **Completeness** (weight 0.3): Coverage of all relevant aspects in the chunk.
- **Relevance** (weight 0.2): Alignment between instruction and response.
- **Structure** (weight 0.1): Clarity of formatting and logical flow.

The composite score is the weighted sum. Records with score < 0.6 trigger a feedback round; records that still fail after $R = 2$ rounds are discarded.

A.8 Latency and Cost Breakdown

Table 6 reports end-to-end latency measured on a single NVIDIA A100 (80 GB) with Qwen2-72B-Instruct 4-bit quantized via Unsloth. Batch size is 1; times are wall-clock averages over 100 records.

Table 6: Latency and Cost Breakdown

Pipeline Stage	Time / Record	Total (35K)	% of Total
Document parsing	—	~2 hrs	4%
Semantic chunking	—	~30 min	1%
Expert selection	0.02 s	~12 min	<1%
Knowledge retrieval	0.15 s	~88 min	3%
Parallel generation	2.8 s	~27 hrs	58%
MetaExpert feedback ($R = 2$)	1.9 s	~18 hrs	38%
Quality scoring	0.05 s	~29 min	1%
Total	~4.9 s	~46 hrs	100%

Notes.

- Document parsing and chunking are I/O-bound and run on CPU; their times depend on PDF complexity.
- Generation and feedback are GPU-bound and dominate the pipeline.
- Parallel collaboration reduces per-record latency by invoking multiple experts in a single forward pass when possible.
- The 46-hour total assumes sequential processing; with 4-way parallel chunk processing across 4 A100 GPUs, the wall-clock time drops to ~12 hours.